

ENGR 102 Summer Bridge**Day 13 Instructor Key**

Loop Design, Break, Continue, and Nested Tracing

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as [and](#), [or](#), and [not](#). Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace skipped and stopped passes.**5 points**

```
score = 1
for value in [3, 0, 4, 8, 2]:
    if value == 0:
        continue
    score = score + value
    if score > 10:
        break
    score = score * 2
print(score)
```

Show score after every reached statement and explain whether 2 is reached.

Answer and explanation

Value 3: score becomes 4, not above 10, then doubles to 8. Value 0 continues with score 8. Value 4 makes score 12, which is above 10, so break runs before doubling. Value 8 and value 2 are never reached. Exact output: 12.

Question 2. Repair the skip/stop choice.**3 points**

```
for reading in [5, -1, 7, 0, 9]:
    if reading < 0:
        break
    if reading == 0:
        break
    print(reading)
```

The rule says negative values are ignored but 0 stops. Give actual output and the minimal repair.

Answer and explanation

Actual output is only 5 because -1 breaks the loop. Replace the first break with continue. Repaired output is 5 and 7; 0 then stops before 9.

Question 3. Program construction**12 points**

Write one complete Python program that satisfies every requirement.

- Repeatedly read integer readings. -1 ends normally.
- A reading of 0 is ignored. A reading above 100 stops immediately and is not counted.
- Positive readings from 1 through 100 are counted and added.
- Print count and total. Include one test that contains both an ignored value and an early-stop value.

Answer and explanation

```
count = 0
total = 0

while True:
    reading = int(input())
    if reading == -1:
        break
    if reading == 0:
        continue
    if reading > 100:
        break
    if reading > 0:
        count = count + 1
        total = total + reading

print(count)
print(total)
```

The normal sentinel is checked before the skip and stop cases. Reading at the top of the loop ensures every continue begins a fresh input pass.

Test 5, 0, 8, 120, 9 prints 2 and 13; 0 is ignored and 9 is never read after 120 stops processing.

Scoring guidance

3 loop/input architecture, 2 sentinel, 2 continue, 2 early stop, 2 count/total/output, 1 discriminating test.